

1. First Come First Serve (FCFS)

C Program :

```
#include <stdio.h>
```

```
int main() {
    int n, i;
    int bt[20], wt[20], tat[20];
    float avg_wt = 0, avg_tat = 0;
    printf("Enter Number of Processes: ");
    scanf("%d", &n);
    for (i = 0; i < n; i++) {
        printf("Enter Burst Time for P%d: ", i + 1);
        scanf("%d", &bt[i]);
    }
    wt[0] = 0;
    for (i = 1; i < n; i++) {
        wt[i] = wt[i - 1] + bt[i - 1];
    }
    for (i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];
        avg_wt += wt[i];
        avg_tat += tat[i];
    }
    printf("\nProcess\tBT\tWT\tTAT\n");
    for (i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\n", i + 1, bt[i], wt[i], tat[i]);
    }

    printf("\nAverage Waiting Time = %.2f", avg_wt / n);
```

```
    printf("\nAverage Turnaround Time = %.2f\n", avg_tat / n);  
    return 0;  
}
```

Shell Program (fcfs.sh) :

```
#!/bin/bash
```

```
echo "Enter Number of Processes"
```

```
read n
```

```
for ((i = 0; i < n; i++))
```

```
do
```

```
    echo "Enter Burst Time for P$((i + 1))"
```

```
    read bt[$i]
```

```
done
```

```
wt[0] = 0
```

```
for ((i = 1; i < n; i++))
```

```
do
```

```
    wt[$i]=$((wt[i - 1] + bt[i - 1]))
```

```
done
```

```
echo
```

```
echo -e "Process\tBT\tWT\tTAT"
```

```
total_wt=0
```

```
total_tat=0
```

```
for ((i = 0; i < n; i++))
```

```
do
```

```
    tat[$i]=$((wt[i] + bt[i]))
```

```
    total_wt=$((total_wt + wt[i]))
```

```
    total_tat=$((total_tat + tat[i]))
```

```
    echo -e "P$((i + 1))\t${bt[i]}\t${wt[i]}\t${tat[i]}"
```

```
done
```

```
avg_wt=$(awk "BEGIN {printf \"%.2f\", $total_wt / $n}")
```

```
avg_tat=$(awk "BEGIN {printf \"%.2f\", $total_tat / $n}")
```

```
echo
```

```
echo "Average Waiting Time = $avg_wt"
```

```
echo "Average Turnaround Time = $avg_tat"
```

Output :

```
(ganesh@LAPTOP-R3RDH9NG)~$ nano fcfs.sh
(ganesh@LAPTOP-R3RDH9NG)~$ chmod +x fcfs.sh
(ganesh@LAPTOP-R3RDH9NG)~$ bash fcfs.sh
Enter Number of Processes
3
Enter Burst Time for P1
12
Enter Burst Time for P2
13
Enter Burst Time for P3
14
fcfs.sh: line 12: wt[0]: command not found

Process BT      WT      TAT
P1      12      12
P2      13      25
P3      14      39

Average Waiting Time = 12.33
Average Turnaround Time = 25.33
```